DermAI

Final Year Project Report

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Certificate of Approval

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This project "**DermAI**" is presented by **Huzaifa bin Shakeel Shaikh, Abdul Wasay, Shahzaib Hussain Pirzada** under the supervision of their project advisor and approved by the project examination committee, and acknowledged by the Hamdard Institute of Engineering and Technology, in the fulfillment of the requirements for the Bachelor degree of Artificial Intelligent.

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Authors' Declaration

We declare that this project report was carried out in accordance with the rules and regulations of Hamdard University. The work is original except where indicated by special references in the text and no part of the report has been submitted for any other degree. The report has not been presented to any other University for examination.

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We, Huzaifa bin Shakeel Shaikh, Abdul Wassay, Shahzaib Pirzaada, solemnly declare that the work presented in the Final Year Project Report titled DermAI has been carried out solely by ourselves with no significant help from any other person except few of those which are duly acknowledged. We confirm that no portion of our report has been plagiarized and any material used in the report from other sources is properly referenced.

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Definition of Terms, Acronyms, and Abbreviations

This section should provide the definitions of all terms, acronyms, and abbreviations required to interpret the terms used in the document properly.

Table 2: Definition of Terms, Acronyms, and Abbreviations

Term	**Description**
DermAI	The proposed AI-based system developed in this project for skin disease detection, classification, and decision support.
CNN (Convolutional Neural Network)	A deep learning architecture designed to automatically learn spatial features from images, widely used in medical image analysis.
Mobile Sam	A state-of-the-art Image Segmentation model used in this project to localize affected skin regions in input images.
ConvNeXt	A modern convolutional neural network architecture inspired by transformer designs, used for skin disease classification.
Object Detection	The process of identifying and localizing specific regions of interest within an image, such as infected skin areas.
Image Classification	A computer vision task that assigns an input image to one of several predefined disease categories.
LLM (Large Language Model)	An advanced language model used to generate disease-specific questions and analyze symptom-based user responses.
Disease-Specific Questioning	An interactive mechanism where follow-up questions are generated based on predicted disease to improve diagnostic accuracy.
Dataset	A structured collection of labeled skin disease images downloaded from Kaggle and used to train and evaluate the models.
Roboflow	An annotation and dataset management tool used for labeling, preprocessing, and augmenting skin disease images.
Transfer Learning	A technique where pre-trained models are fine-tuned on a new dataset to improve performance with limited data.
Doctor Consultation Module	A system component that allows users to seek professional consultation after automated diagnosis.

Abstract

Skin diseases are among the most common health problems worldwide, and their early and accurate diagnosis remains challenging due to visual similarity between conditions, variations in skin tone, and limited access to dermatologists. This project presents **DermAI**, an intelligent skin disease analysis system that integrates deep learning–based image processing with disease-specific interactive questioning and clinical consultation support.

The proposed system follows a multi-stage diagnostic pipeline. Initially, **Mobile Sam** is employed for object detection to accurately localize affected skin regions within uploaded images. The detected region is then passed to a **ConvNeXt-based image classification model**, which predicts the most probable skin disease category. Based on the predicted disease, a **disease-specific Large Language Model (LLM)** dynamically generates targeted questions to collect relevant symptom information from the user. This hybrid approach reduces reliance on image data alone and improves diagnostic reliability by incorporating contextual symptom responses.

All datasets used in this project are carefully annotated using **Roboflow**, ensuring consistent labeling and preprocessing. The final diagnosis is generated through a decision fusion mechanism that combines visual classification results with symptom-based analysis. Once the diagnosis is established, the system enables **doctor consultation**, allowing medical professionals to review the results and provide expert guidance.

The system is designed using the **Waterfall development model**, ensuring structured requirement analysis, design validation, implementation, and testing. DermAI aims to assist users in preliminary skin disease assessment, reduce diagnostic delays, and support dermatologists by providing AI-assisted insights rather than replacing medical judgment. The results demonstrate that integrating object detection, modern CNN architectures, and LLM-driven interaction significantly enhances diagnostic accuracy and system usability.

Keywords:

1. Skin Disease Detection
2. Mobile Sam
3. ConvNeXt
4. Medical Image Analysis
5. Convulational Neural Network (CNN)
6. Large Language Model (LLM)
7. Computer-Aided Diagnosis
8. Deep Learning
9. Roboflow Annotation Pipeline
10. Telemedicine Consultation System

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CHAPTER 1

INTRODUCTION

1.1 Motivation

Skin diseases are among the most common health problems worldwide, yet timely and accurate diagnosis remains a challenge, especially in regions with limited access to dermatologists. Many patients rely on self-assessment or delayed consultations, which can lead to disease progression and improper treatment. The motivation behind this project is to address this gap by leveraging modern artificial intelligence techniques to assist in early, accessible, and reliable skin disease assessment.

Recent advancements in computer vision and deep learning have shown strong potential in medical image analysis. Models such as Mobile Sam and ConvNeXt enable efficient object detection and high-accuracy image classification, making them suitable for real-time medical applications. Additionally, the integration of disease-specific questioning through Large Language Models (LLMs) allows the system to mimic a dermatologist's diagnostic reasoning by combining visual evidence with patient-reported symptoms.

Another key motivation is to reduce dependency on manual processes and subjective judgment by providing a standardized, AI-driven decision-support tool. By using Roboflow for precise image annotation and structured datasets, the system aims to improve model reliability and generalization across different skin tones and image conditions. Furthermore, the inclusion of doctor consultation as a final step ensures that the system acts as an assistive tool rather than a replacement for medical professionals, maintaining ethical and clinical responsibility.

Overall, this project is motivated by the need for an intelligent, scalable, and user-friendly dermatology assistance system that enhances early detection, improves diagnostic confidence, and supports healthcare delivery through the effective use of AI technologies.

1.2 Problem Statement

Skin diseases are among the most common health issues worldwide, yet timely and accurate diagnosis remains a challenge, especially in regions with limited access to dermatologists. Many individuals rely on self-assessment or online information, which is often inaccurate, unverified, or misleading. Existing AI-based dermatology systems mostly focus on image-only analysis, ignoring critical symptom-related information such as itching, pain, duration, or spread, which dermatologists routinely consider during diagnosis. Additionally, most available solutions are either expensive, non-interactive, or limited to detecting only a few severe conditions, making them unsuitable for early-stage and common skin disease identification.

Another major limitation is the lack of an integrated workflow that mimics real clinical practice. Current systems rarely combine object detection, disease classification, intelligent symptom-based questioning, and professional consultation within a single platform. Variations in image quality, lighting conditions, and skin tones further reduce diagnostic reliability when visual data is used alone. Therefore, there is a clear need for an intelligent, interactive, and end-to-end dermatology support system that can accurately detect skin conditions from images, refine predictions through disease-specific questioning, and seamlessly guide users toward medical consultation when required.

The problem addressed in this project is the design and development of an AI-powered dermatological assistance system that integrates object detection, image classification, and language-based symptom analysis to improve diagnostic accuracy and accessibility, while maintaining scalability, usability, and clinical relevance.

1.3 Goals and Objectives

The goal of this project is to develop an AI-powered dermatological screening system that provides reliable preliminary identification of common skin diseases through automated image analysis and intelligent symptom questioning. The system aims to detect and localize affected skin regions using Mobile sam for image segmentation, classify diseases accurately using a modern deep learning model such as ConvNeXt, and enhance diagnostic confidence through disease-specific follow-up questions generated by a Large Language Model (LLM). By combining visual features with contextual symptom information, the project seeks to reduce misclassification and improve decision reliability. Additional objectives include designing a scalable and modular architecture for future model upgrades, ensuring user data privacy and ethical compliance, supporting both desktop and mobile platforms, and offering an accessible solution that assists users—particularly in underserved areas—while clearly complementing, not replacing, professional medical consultation.

1.4 Project Scope

The scope of this project covers the design and development of an AI-based dermatological assistance system that performs automated skin disease analysis and preliminary diagnosis. The system focuses on detecting affected skin regions using the Mobile Sam for image segmentation model, followed by disease classification through a deep learning classifier such as ConvNeXt. Based on the initial classification results, the system generates disease-specific follow-up questions using a Large Language Model (LLM) to collect contextual symptom information that cannot be determined from images alone. The project also includes integrating a doctor consultation module to allow users to seek professional guidance after receiving the AI-generated assessment. The application is designed to operate on desktop and mobile platforms with an English-language interface and online functionality. However, the system does not aim to replace certified medical professionals, provide final clinical diagnoses, prescribe medication, or handle emergency medical cases. Instead, it is intended solely for early-stage screening and decision support while ensuring data privacy, ethical compliance, and modularity for future enhancements.

CHAPTER 2

RELEVANT BACKGROUND & DEFINITIONS

This chapter provides the essential background knowledge and core concepts required to understand the development of the DermAI system. It introduces the key technologies and methodologies used in the project, including medical image analysis, deep learning, and artificial intelligence. Fundamental concepts related to skin disease diagnosis, object detection, image classification, and natural language processing are briefly explained to establish the technical foundation of the system.

The chapter also outlines the deep learning models and tools employed in DermAI, such as mobile sam for image segmentation, ConvNeXt for image classification, and Roboflow for dataset annotation. Additionally, the role of Large Language Models (LLMs) in disease-specific questioning and decision support is discussed to highlight how contextual symptom information improves diagnostic accuracy. Relevant definitions and terminology are included to ensure clarity and consistency throughout the report.

2.1 Background of Skin Diseases

Skin diseases are among the most common health problems worldwide and can affect people of all ages. Conditions such as acne, eczema, fungal infections, psoriasis, and skin cancer often require early diagnosis to prevent complications. However, access to qualified dermatologists is limited in many regions, especially in rural and underdeveloped areas. Diagnosis of skin diseases is primarily visual and depends on the experience of medical professionals, which can lead to human error or delayed treatment. These challenges highlight the need for automated, technology-driven systems that can assist in early screening and preliminary diagnosis of skin conditions.

Table 3.**1**

Aspect	**Traditional Diagnosis**	**DermAI (Proposed System)**
Diagnosis Method	Manual visual inspection by dermatologist	AI-based image analysis using deep learning
Accessibility	Limited, especially in rural or underdeveloped areas	Easily accessible through mobile and desktop platforms
Time Required	Time-consuming due to appointments and waiting	Provides instant preliminary results
Cost	High consultation and travel costs	Low-cost or free initial screening
Accuracy	Depends on doctor’s experience and human judgment	Consistent predictions using trained AI models
Scalability	Cannot serve large populations simultaneously	Highly scalable and can serve multiple users at once
Early Detection	May be delayed due to lack of access	Enables early-stage screening and awareness
User Interaction	Limited to in-person consultation	Interactive disease-specific questioning using AI
Error Handling	Prone to human fatigue and subjectivity	Reduced bias through standardized model inference
Follow-up Support	Requires physical visits	Supports digital doctor consultation module

2.2 Artificial Intelligence in Healthcare

Artificial Intelligence (AI) has emerged as a powerful tool in modern healthcare by enabling automated analysis, prediction, and decision support. In medical imaging, AI techniques such as deep learning and computer vision are widely used for detecting and classifying diseases with high accuracy. AI-based systems can analyze large volumes of data, identify patterns that may not be visible to humans, and provide fast and consistent results. In dermatology, AI helps improve diagnostic efficiency, reduce workload on doctors, and support early disease detection, making healthcare more accessible and cost-effective.

2.3 (Mobile SAM)

Object segmentation is a computer vision task that involves identifying and precisely outlining regions of interest within an image at the pixel level. Mobile SAM (Segment Anything Model for Mobile) is a lightweight and efficient segmentation model designed to perform accurate segmentation on resource-constrained devices. In this project, Mobile SAM is used to segment the affected skin regions from input images before classification. By accurately isolating the lesion area, the system minimizes background interference and ensures that only clinically relevant regions are passed to the classification model. Mobile SAM was selected due to its optimized architecture, fast inference speed, and ability to deliver high-quality segmentation results while remaining suitable for mobile and real-world deployment scenarios.

2.4 Image Classification (ConvNeXt)

Image classification is the process of assigning a label to an image based on its visual content. ConvNeXt is a modern convolutional neural network architecture inspired by Transformer-based designs while maintaining the efficiency of CNNs. In this project, ConvNeXt is used to classify detected skin lesions into different disease categories. The model provides strong performance with fewer parameters and better feature representation compared to traditional CNNs. Its design makes it suitable for medical image classification tasks where accuracy and generalization are critical.

2.5 Large Language Models in Medical Systems

Large Language Models (LLMs) are advanced AI models capable of understanding and generating human-like text. In DermAI, LLMs are used after image classification to generate disease-specific questions for the user. These models help collect symptom-related information such as itching, pain, duration, and medical history. By combining visual predictions with textual symptom analysis, the system mimics real clinical reasoning. LLMs also enhance user interaction by providing natural, conversational, and context-aware questioning, making the system more intelligent and user-friendly.

2.6 Definitions

- **Artificial Intelligence (AI):** The ability of machines to simulate human intelligence, including learning, reasoning, and decision-making.
- **Deep Learning:** A subset of machine learning that uses neural networks with multiple layers to learn complex patterns from data.
- **Object Detection:** A technique used to identify and locate objects within an image.
- **Image Classification:** The task of assigning a category label to an image based on its content.
- **Mobile Sam:** A real-time image segmentation model used to detect skin lesion regions in images.
- **ConvNeXt:** A convolutional neural network architecture used for image classification.

- **Large Language Model (LLM):** A model trained on large text datasets to understand and generate natural language.
- **DermAI:** The proposed AI-based skin disease detection and consultation system.
- **Roboflow:** An online platform used for dataset annotation, preprocessing, and management.
- **Lesion:** An abnormal area of skin that may indicate disease or infection.

CHAPTER 3

LITERATURE REVIEW & RELATED WORK

3.1 Literature Review

Recent research in medical image analysis shows that deep learning techniques have become highly effective for automated skin disease detection and classification. Publicly available dermatology datasets hosted on platforms such as Kaggle have enabled researchers to train robust models for identifying common skin conditions using real-world images. Object detection models, particularly Mobile Sam-based architectures, are widely used to localize affected skin regions due to their real-time performance and high detection accuracy; newer versions such as Mobile Sam further enhance detection efficiency through improved feature extraction and optimized training pipelines. For classification tasks, modern convolutional neural networks like ConvNeXt have demonstrated superior performance over traditional CNNs by incorporating transformer-inspired design principles while retaining convolutional efficiency, making them well-suited for fine-grained skin disease classification. Additionally, dataset annotation and preprocessing tools such as Roboflow are commonly used to improve data quality, consistency, and augmentation, which significantly contributes to model generalization. Recent studies also highlight that combining visual analysis with contextual reasoning enhances diagnostic reliability, motivating the integration of intelligent decision-support mechanisms. Building upon these advancements, this project adopts Mobile Sam for lesion detection, ConvNeXt for disease classification, and Roboflow for dataset annotation using a Kaggle-based skin disease dataset, forming a unified framework for accurate and efficient skin disease analysis.

3.2 Related Work

Recent research in automated skin disease diagnosis has increasingly focused on deep learning–based computer vision techniques to assist dermatological screening. Several studies have demonstrated that convolutional neural networks and modern architectures can effectively learn visual patterns from skin images and achieve promising classification accuracy. Object detection models such as Mobile Sam have been widely adopted to localize affected skin regions before classification, improving robustness by focusing on relevant areas of the image. At the same time, transformer-inspired architectures like ConvNeXt have gained attention due to their strong feature extraction capabilities and improved performance over traditional CNNs in medical image analysis. Publicly available datasets hosted on platforms such as Kaggle have enabled researchers to train and evaluate these models in controlled settings. However, most existing works rely on image-based prediction alone and do not incorporate interactive or contextual analysis. Building on these prior studies, this project utilizes a Kaggle-based skin disease image dataset along with Mobile Sam for image segmentation and ConvNeXt for classification, aiming to create a more accurate and practical AI-driven dermatology support system.

3.3 Gap Analysis

1-Most existing systems rely only on image classification without performing object detection to localize affected skin regions.
2-Background noise and irrelevant visual information in images often reduce classification accuracy.
2-Many prior studies use traditional CNN architectures instead of modern models like ConvNeXt.
4-Limited exploration of lightweight yet high-performing architectures suitable for real-world deployment.
5-Lack of integration between object detection and classification within a unified pipeline.

6-Absence of disease-specific questioning to support visual diagnosis.
7-Existing systems largely ignore contextual symptom information provided by users.
8-Minimal focus on interactive user engagement during the diagnostic process.
9-Poor scalability and modular design in many research prototypes.
10-Limited support for post-diagnosis workflows such as expert or doctor consultation.

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